

Software Developer Intern

Technical Assignment

Project: Software Quotation Management System

Maximum Time: 48 Hours

Purpose: Build a simple quotation management application for a software/finance software company.

1. Assignment Objective

Develop a functional web application that allows an authenticated user to create, calculate, save, view, and delete software quotations. The application should demonstrate practical knowledge of frontend development, backend/database integration, form handling, business logic, and deployment.

2. Required Technology Stack

- **Frontend:** React.js or Next.js
- **Backend & Database:** Supabase
- **Authentication:** Supabase Authentication
- **Deployment:** Vercel
- **Language:** JavaScript or TypeScript
- **Version Control:** Git & GitHub
- Tailwind CSS, Bootstrap, Material UI, or another UI library may be used for styling.

3. Login

Create a simple login page using Supabase Authentication.

- Email
- Password
- Login button
- Logout functionality
- Redirect the user to the quotation management page after successful login

4. Quotation Management

After login, the user should be able to create and manage quotations.

- Create a new quotation
- View previously created quotations
- View an individual quotation
- Delete a quotation

5. Create Quotation Form

Customer Information

- Customer Name
- Company Name
- Email

- Phone

Quotation Information

- Quotation Number
- Quotation Date
- Valid Until

Product / Service Information

The user should be able to add one or more products/services. Each row should contain:

Field	Example
Product Name	Accounting Software
Quantity	2
Unit Price	■25,000
Discount	5%

Provide an + **Add Product** button and allow the user to remove product rows.

6. Automatic Calculation

The quotation amount must be calculated automatically whenever quantity, price, or discount changes.

Calculation	Formula
Gross Amount	Quantity × Unit Price
Discount Amount	Gross Amount × Discount %
Net Amount	Gross Amount – Discount Amount
Subtotal	Sum of all Net Amounts
GST	Subtotal × GST %
Grand Total	Subtotal + GST

Example Calculation

Product: Accounting Software

Quantity: 2

Unit Price: ■25,000

Discount: 5%

GST: 18%

Gross Amount: ■50,000

Discount: ■2,500

Subtotal: ■47,500

GST: ■8,550

Grand Total: ■56,050

7. Save Quotation in Supabase

When the user clicks **Save Quotation**, the quotation and its items must be stored in Supabase.

Suggested Database Structure

quotations	quotation_items
id	id
quotation_number	quotation_id
customer_name	product_name
company_name	quantity
email	unit_price
phone	discount
quotation_date	amount
valid_until	
subtotal	
gst	
total	
created_at	

You may use a different database structure if you can explain your design.

8. Quotation List

Display previously created quotations in a table. The table should include at least:

- Quotation Number
- Customer
- Amount
- Date
- View
- Delete

9. View Quotation

When the user clicks **View**, display the quotation in a clean and professional format containing customer details, quotation details, products, quantity, price, discount, subtotal, GST, and grand total.

10. Basic Validation

- Customer name is required.
- Email should be valid.
- Quantity must be greater than 0.
- Price cannot be negative.
- At least one product is required.
- Quotation date is required.

11. UI/UX Requirements

- Clean and professional interface
- Responsive layout

- Clear buttons and form labels
- Readable quotation table
- Loading state where appropriate
- User-friendly error messages

12. GitHub Requirements

Create a GitHub repository and use Git during development. The repository should contain the source code, README.md, and an .env.example file.

Do not commit actual passwords, private keys, Supabase service-role keys, or other sensitive credentials.

13. Deployment

Deploy the completed application to Vercel. The submitted application must be accessible through the live deployment URL.

14. README Requirements

- Project overview
- Features
- Technology stack
- Installation instructions
- Required environment variables
- Supabase setup
- Deployment information

15. Optional Bonus Features

- Edit quotation
- Search quotations
- Filter quotations
- PDF generation
- Print quotation
- GST selection
- Basic dashboard

16. Time Limit

Maximum completion time: 48 hours.

The 48-hour period begins from the time the assignment is shared. Candidates should prioritize the core requirements before attempting optional features.

17. Final Submission

Item	Required
GitHub Repository	Yes
Live Vercel URL	Yes

Test Login Credentials	Yes
README.md	Yes

Expected Result: Login → Create Quotation → Add Products → Automatically Calculate Total → Save to Supabase
→ View Quotations → View Individual Quotation → Delete Quotation → Deploy on Vercel.